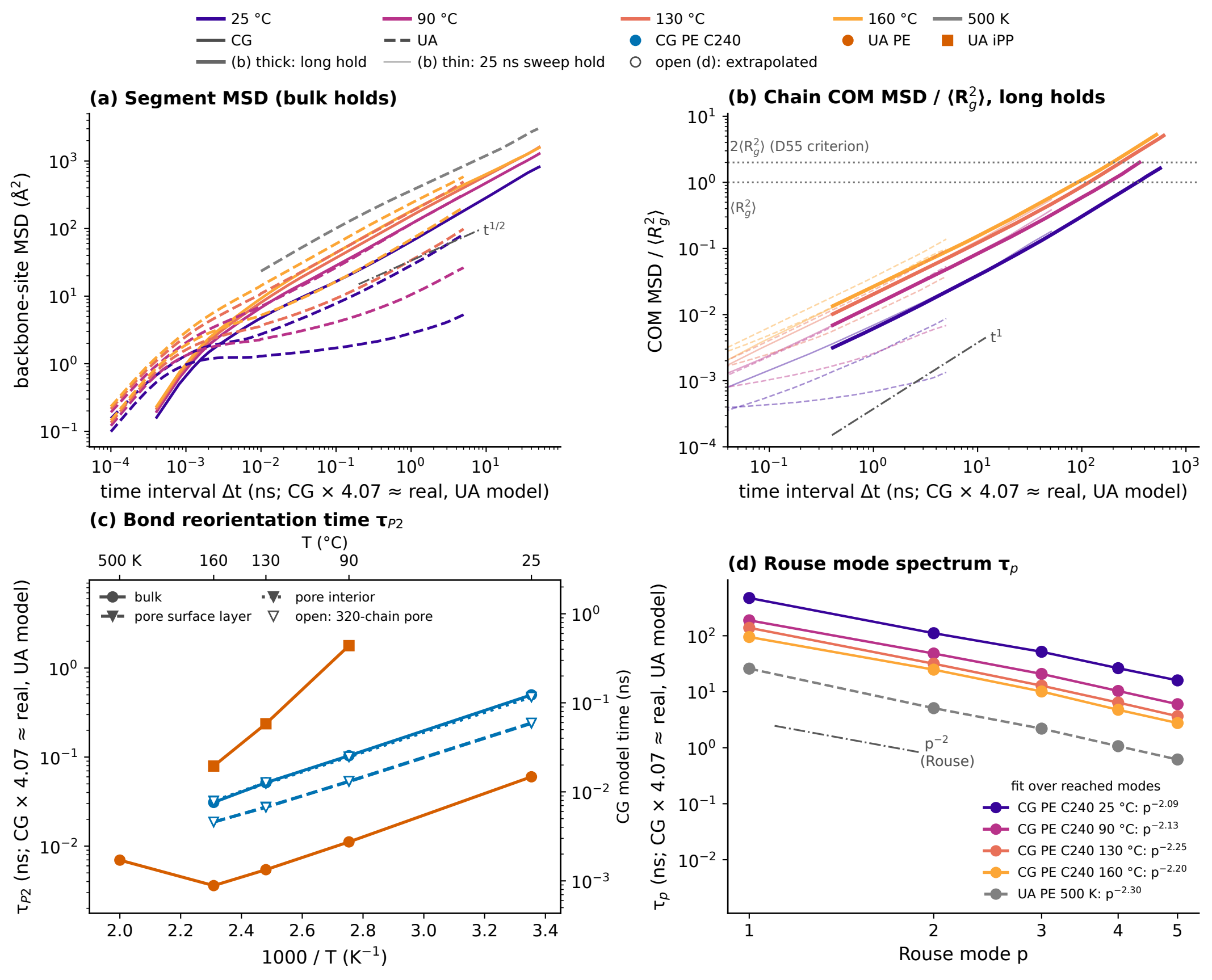

25 °C
90 °C
130 °C
160 °C
500 K
CG
UA
CG PE C240
UA PE
UA iPP
(b) thick: long hold
(b) thin: 25 ns sweep hold
open (d): extrapolated
(a) Segment MSD (bulk holds)
backbone-site MSD (Å²)
t^1/2
time interval Δt (ns; CG × 4.07 ≈ real, UA model)
(b) Chain COM MSD / ⟨R_g²⟩, long holds
2⟨R_g²⟩ (D55 criterion)
⟨R_g²⟩
COM MSD / ⟨R_g²⟩
t^1
time interval Δt (ns; CG × 4.07 ≈ real, UA model)
(c) Bond reorientation time τ_P2
T (°C)
500 K
160
130
90
25
bulk
pore surface layer
pore interior
open: 320-chain pore
τ_P2 (ns; CG × 4.07 ≈ real, UA model)
CG model time (ns)
1000 / T (K^-1)
(d) Rouse mode spectrum τ_p
τ_p (ns; CG × 4.07 ≈ real, UA model)
p^-2
(Rouse)
fit over reached modes
CG PE C240 25 °C: p^-2.09
CG PE C240 90 °C: p^-2.13
CG PE C240 130 °C: p^-2.25
CG PE C240 160 °C: p^-2.20
UA PE 500 K: p^-2.30
Rouse mode p